Week 2 – Modeling Property Scenarios: Idealized Answers

Geophysics 3A: Potential Fields
& Geothermics

These model answers illustrate one strong, defensible pathway for each scenario. They reflect strategies discussed in the Week 2 notes on physical properties: direct measurement; composition/mineral-mode estimates; geochemical proxies (e.g., K–U–Th for heat production); empirical correlations (e.g., density–velocity); thermodynamic/ P–T corrections; and cautious use of indirect geophysical inference where appropriate. (Course notes on physical properties and estimation strategies).[1]

Scenario A: Basin Density Model (Resources/Tectonics)

Goal recap. Build a depth-dependent density model for a layered basin to support gravity modelling and compaction analysis.

Questions for the geologist (2–3)

- What are the mapped stratigraphic units and their lateral continuity; where do cores tie to the seismic horizons? (Needed to honour layering and compaction trends.) (Notes: properties vary by lithology and state variables.)
- What are porosity/ grain-density measurements from cores (dry, saturated, submerged) and how do they vary with depth/diagenesis? (Porosity affects bulk density; cracks/porosity can matter even when crystalline rock density changes little.)
- Evidence for fluid substitution, cementation, or overpressure that would alter compaction laws?

Preferred method & why
Core-calibrated density–depth curves per facies, tied to seismic stratigraphy. Compute bulk density from core measurements (grain density + porosity), derive empirical $\rho(z)$ trends by unit, then propagate along seismic horizons; fill gaps with composition/lithology-based estimates where no core exists. Validate forward-modelled gravity against observations and iterate. This leverages direct measurements where available and honours the layered geometry the seismic defines, while acknowledging that density is largely bulk-additive and predictable from composition and porosity.

Pros

- Uses *direct* core data to anchor $\rho(z)$; unit-specific compaction captured.
- Seismic ties enforce stratigraphic consistency across the basin.

Cons

- Core coverage may be sparse; lateral heterogeneity between wells remains.
- Fluid/pressure effects and diagenetic changes introduce uncertainty in porosity–depth laws.

[1] See course notes: state-variable control (P, T, composition, porosity/cracks), mixing-behaviour by property class, composition-based prediction of A, k, and density trends; limitations and non-uniqueness of proxy methods.

Scenario B: Mountain-Belt Cross-Section (Tectonics)

Goal recap. Provide unit densities for a balanced cross-section (dam siting context), testing isostasy and gravity fit with sparse samples.

Questions for the geologist

- What are the dominant lithologies and metamorphic grades along section (map + prior exploration reports)? (Density correlates with composition and metamorphic state.)
- Are there systematic fabrics (foliation, layering) or fluid alteration zones that might affect cracks/porosity?
- Availability of any archival core/hand samples for grain-density validation?

Preferred method & why
Composition/lithology-based density estimates by unit, calibrated with any available samples, with P–T adjustments from metamorphic grade. Use mapped lithologies and representative compositions to assign grain densities, adjust for metamorphic state (denser eclogitic/mafic assemblages, lower for felsic), and apply modest porosity where appropriate (sediments). Validate against regional gravity; iterate unit densities within plausible ranges to achieve fit. Composition-based density is supported by the notes (e.g., felsic/mafic trends, Fe–Mg control), while acknowledging uncertainties from fabric and fractures.

Pros

- Feasible with sparse sampling; leverages widely available geological maps.
- Naturally integrates with gravity modelling for plausibility checks.

Cons

- Non-uniqueness; multiple density assignments can fit gravity.
- Localized alteration/cracking not captured by map-scale lithology alone.

Scenario C: Upper-Mantle Density for Isostasy (Geodynamics)

Goal recap. Convert regional tomography (V_p, V_s) into an upper-mantle density model for isostasy and subduction dynamics (to 410 km).

Questions for the geologist

- What petrologic constraints exist for wedge and slab compositions (peridotite, pyrolite variants, hydrated/ metasomatized domains)? (Composition controls density and elastic moduli.)
- Thermal structure priors (slab geotherm, wedge temperatures) and presence of partial melt or water? (Temperature decreases velocity and density via thermal expansion/softening; hydration affects velocities.)
- Expected phase boundaries (e.g., 410 km olivine–wadsleyite) along the transect?

Preferred method & why
Velocity-to-density conversion using mineral-physics-informed relations with P–T–composition corrections. Start from tomography to obtain V_p, V_s anomalies; adopt a reference mantle model and compute density via calibrated $\rho(V_p, V_s, P, T, X)$ relations (or ρ from K_S, μ, and thermal expansion), applying geotherm-based temperature corrections and composition scenarios. This aligns with notes that properties depend on P–T and composition; velocities increase with pressure, decrease with temperature, and cracks/fluids complicate shallow mantle.

Pros

- Directly leverages available tomography; consistent with geodynamic inputs.
- Can explore scenario envelopes (dry vs hydrated wedge; warm vs cold slab).

Cons

- Conversion non-unique; requires mineral-physics assumptions and geotherm priors.
- Attenuation, anisotropy, and melt can bias V_p/V_s without straightforward density equivalents.

Scenario D: Aeromagnetic Susceptibility Model (Mineral/Exploration)

Goal recap. Build a susceptibility model that distinguishes prospective units and identifies where remanence may invalidate induced-only assumptions.

Questions for the geologist

- Lithologies and expected magnetic minerals (magnetite/hematite/ilmenite) and their alteration/oxidation states? (Magnetic properties are mineralogically controlled.)
- Evidence for remanent magnetization (cooling, chemical remanence, volcanic units)? Domains where induced-only is unsafe?
- Any existing hand-sample/ core susceptibility measurements to calibrate ranges by unit?

Preferred method & why

Hybrid approach: direct susceptibility measurements on representative samples to anchor unit-wise χ, supplemented by lithology-based ranges elsewhere; test for remanence during inversion/interpretation. Susceptibility is best constrained by measurement because it depends on specific magnetic phases and grain size; where sampling is impossible, use typical χ ranges by lithology, flagged with higher uncertainty. During modeling, evaluate phase/ directional misfits suggestive of significant remanence. This mirrors notes emphasizing mineralogical control and limits of indirect proxies.

Pros

- Anchors inversion with real χ values; improves discrimination of units.
- Explicitly surfaces remanence risk where induced models fail to match data.

Cons

- Sampling bias (outcrops/roads) may under-represent subsurface units.
- χ alone does not quantify remanence; requires additional magnetic information or assumptions.

Scenario E: Heat-Flow Study in Granitic Terrane (Geothermal/Environmental)

Goal recap. Estimate $k(T)$ and radiogenic heat production for granites and metasediments to forecast geotherms and transients; pilot holes exist in overlying basin, granite known only from gravity.

Questions for the geologist

- Representative lithologies/compositions of granites and metasediments (regional analogues if direct samples absent)? (Composition controls A and influences k.)
- Any K–U–Th data from nearby outcrops/archives; variability expected across facies?
- Basin geometry and basement depth from gravity/seismic to set layer thicknesses for thermal models?

Preferred method & why

Composition-based A from K–U–Th where available; conductivity $k(T)$ estimated from composition/mineralogy (or V_p proxy in basin sediments), constrained by pilot-hole temperature gradients; basement geometry from gravity. Use pilot wells to calibrate basin sediment k and surface heat flow; estimate granite A from regional geochemistry and map plausible k from mineralogy; propagate uncertainties with envelope models. Notes indicate heat production can be predicted from composition and that conductivity often correlates with composition or V_p; tie to layer geometry for thermal modeling.

Pros

- Leverages existing pilot-hole data to constrain near-surface thermal state.
- Physically motivated estimates of A and k from composition/mineralogy.

Cons

- Basement properties remain uncertain without direct samples; multiple granite variants possible.
- k anisotropy and temperature dependence add uncertainty in transient predictions.

Scenario F: Geothermal Heat Flow at Base of Antarctic Ice Sheet (Geothermal/Glaciology)

Goal recap. Develop basal heat-flow inputs for ice-dynamics stability assessment; nunatak samples/geochemistry exist near coast; regional seismic (low-res) and gravity (high-res) are available; interior is poorly exposed.

Questions for the geologist

- Crustal/lithologic provinces expected beneath the ice interior (from regional geology and nunatak analogues)? (Composition controls A and influences k.)
- Any heat-flow or temperature constraints (boreholes, radar internal layers/ basal conditions) to calibrate models? (External geophysics to cross-check.)
- Evidence for radiogenic granites vs mafic provinces inland; degree of heterogeneity?

Preferred method & why

Regional prior approach: map A and k using composition inferences from nunatak geochemistry and tectonic province analogues; constrain crustal thickness/geometry with gravity + seismic; assimilate ice-temperature/radar indicators where available. Predict A from K–U–Th of coastal outcrops with uncertainty bounds; extrapolate cautiously inland using tectonic/domain mapping; assign k from mineralogy/rock type ranges; use gravity+seismic to define thermal layer thicknesses; test model families against ice temperature/flow indicators. Notes support composition-based A and k estimation and stress uncertainty where sampling is limited.

Pros

- Feasible despite limited exposure; integrates multiple datasets for consistency.
- Produces uncertainty-bounded priors suitable for ice-flow modelling sensitivity tests.

Cons

- Extrapolation from nunataks may misrepresent inland lithologies; large uncertainty in A.
- Sparse calibration for $k(T)$ and basal conditions; anisotropy/fluids may not be captured.